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Software Engineering

Req. Modelling

COMP1003 - Lecture 5

Dr Horia A. Maior

Last Lecture Learning Outcomes

- The difference between requirements and specifications
- The difference between functional vs non-functional (req & spec)
- Converting data from Req Elicitation into actual requirements
- How to go about checking/validating your ‘actual requirements’

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Recap on our situation

- So you've spent hours interviewing people, surveying, observing people in practice, trying it yourself
- How do you put 'that' into a Requirements Document?
- You need to produce models of what you saw

Requirements Engineering

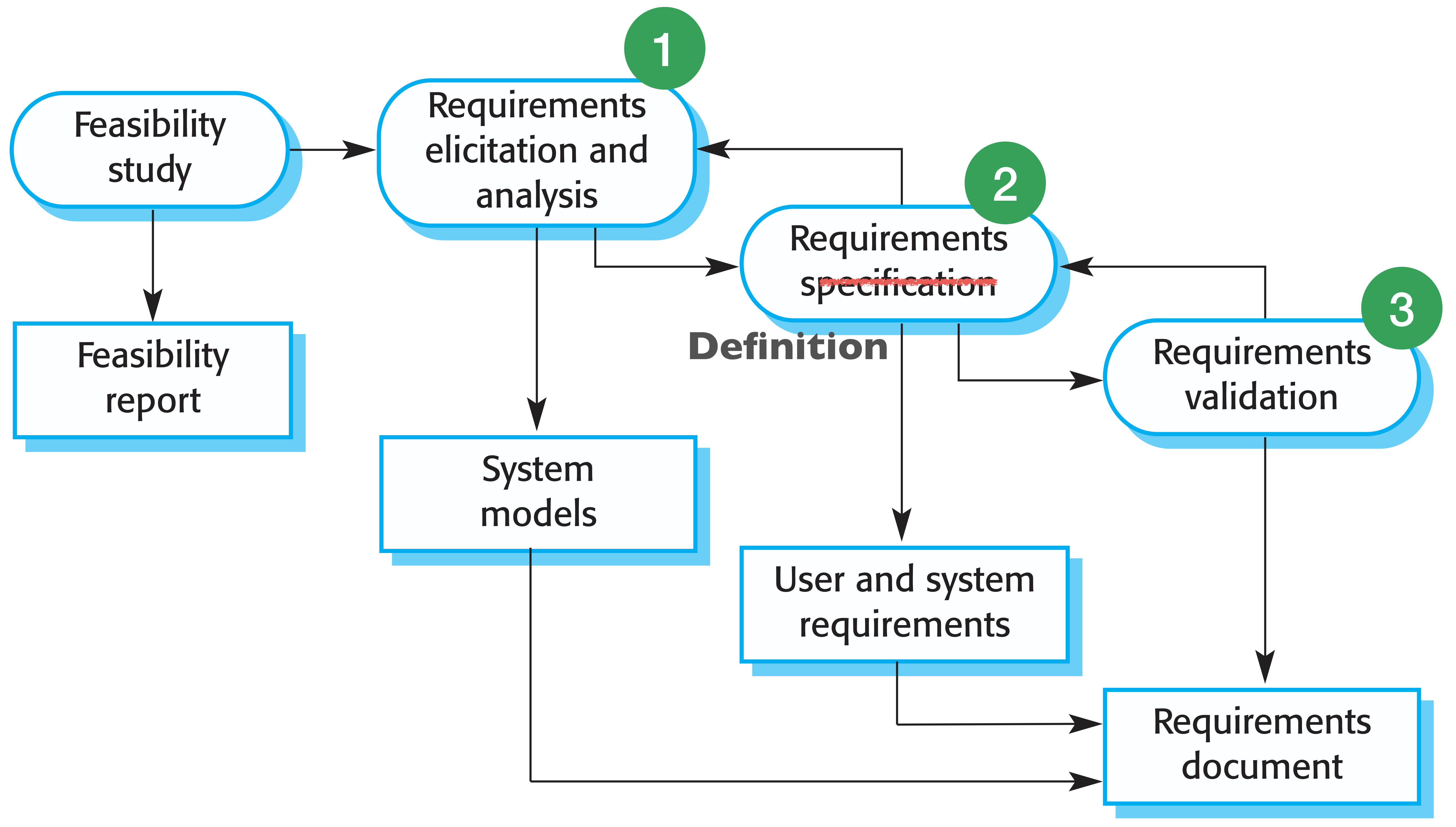
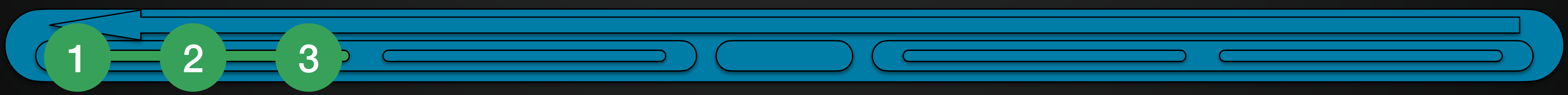


Figure 2.4 The requirements engineering process



Requirements Engineering

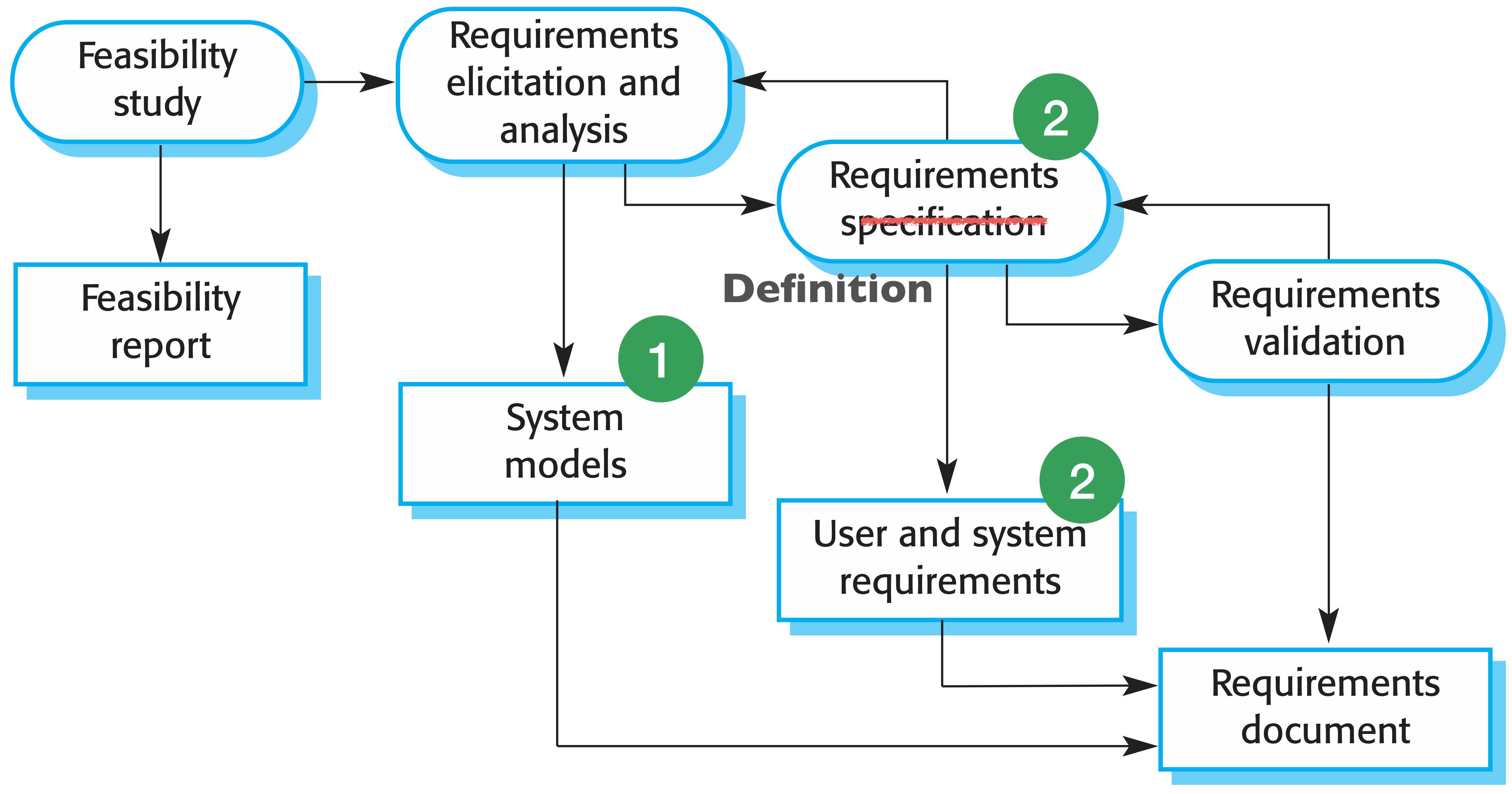
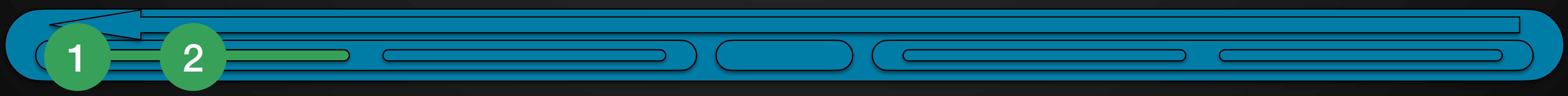


Figure 2.4 The requirements engineering process



Today's Learning Outcomes

- 1) What requirements models are, and why they are useful
- 2) Different types of UML diagrams
- 3) Actually, models aren't just for requirements
  - this is a useful transition point to our next stage

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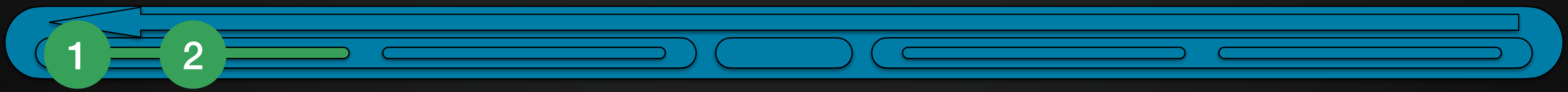
What are requirements models for?

Requirements Models

“work models help you take the next leap in understanding - from knowing the **work of individual** stakeholders to understanding the fundamental structure of **work for a whole stakeholder population**.”

– Beyer & Holtzblatt, Contextual Design

- **Synthesising** all the requirements you collected into key requirements
- You start making diagrams that represent **how requirements relate**
- Then you have a **comprehensive set** of **integrated requirements**



Requirements Models

- Representations of how something works
- Different representations convey different useful benefits
- Models are **selective** representations to portray an **important aspect**

"The most important aspect [to remember] is that it **[has to leave] out detail**. A model is an abstraction of the [process] being studied rather than an alternative representation of that [process] … **An abstraction deliberately simplifies** and picks out the most salient characteristics."

– Ian Sommerville (2011) Chapter 5, Systems Modeling

Just Requirements? No

- Sommerville (the book) outlines a number of models:
  - Context models, Process models, Interaction models, etc
- But actually, we've been using *some* models already
  - and there are plenty of others

Models are used during the requirements engineering process to help
1) derive the requirements for a system,
2) during the design process to describe the system to engineers implementing the system, and
3) after implementation **to document the system**s structure and operation.

– Ian Sommerville (2011) Chapter 5, Systems Modeling

Important Thought from the Lab

- Did you do a use case diagram for their old process?
  - That would be requirements
- Did you do a use case diagram for the new system?
  - That would be specifications
- Like choosing a model, you should be clear what you are doing
- e.g. If the coursework asks you to create diagram X for ‘requirements’ - then make sure you do it for requirements, and not spec

Most Important Thing:

- 1) What do you want to show?
  - a complex decision process
  - a strict sequence of actions
  - subtasks in a complex use case
  - other software to export to, import from, use
  - the structure of a system, etc
- 2) Which diagram or model is best for this?
- Avoid ‘trying to show everything in one diagram’

Context Diagrams

Context Diagrams

- During Req Gathering, you identified a number of systems you will have to interact with
  - authentication systems, bluecastle, finance, etc
- You want to show how all these systems inter-relate
  - Use Case Diagrams could have an ‘actor’ - but thats not what UCDs are ‘for’
  - You could use a flow diagram (Activity Diagram - see later) - but thats not what they are ‘for’
- Context Models are especially **for this problem**

Context Diagrams

- Shows related systems
- These are often part of the Non-Functional requirements
- Can be simple like this one
- Or more detailed (next slide)
  - showing what is transferred etc
  - or conditions of when/why
- Even show components of a multi-part system (twitter)

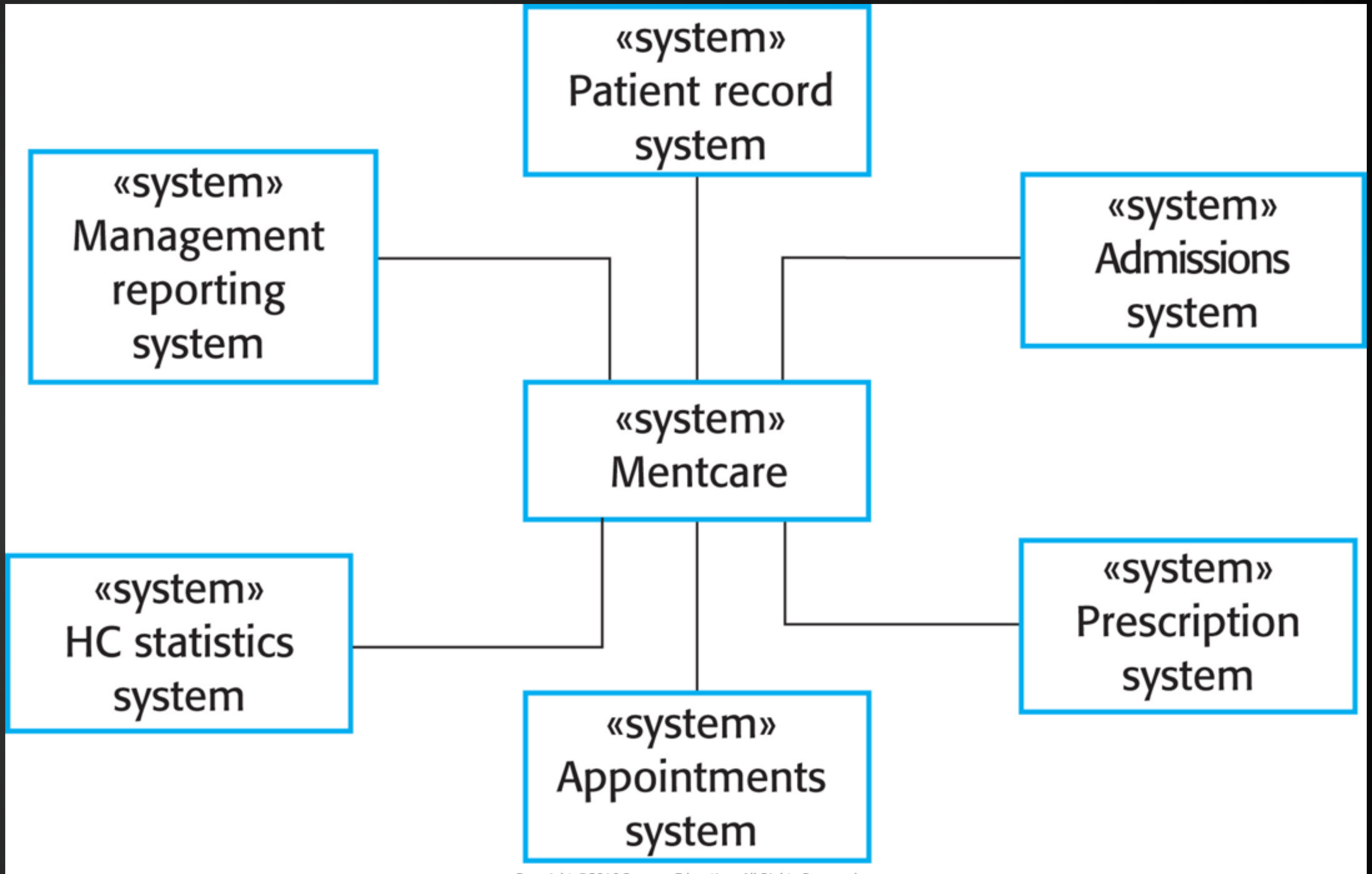


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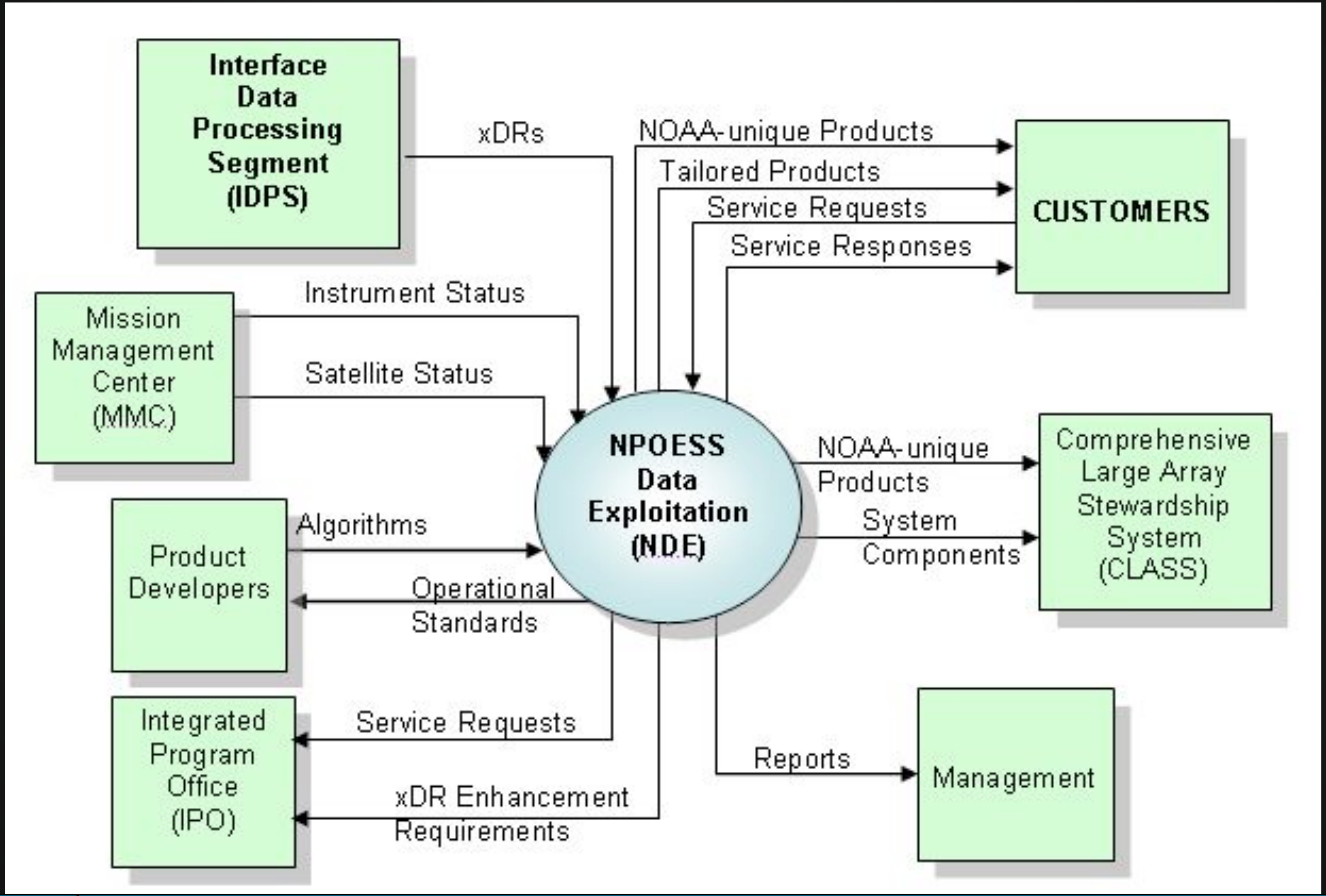
Figure 5.1 *The context of the Mentcare system*

Context Diagrams



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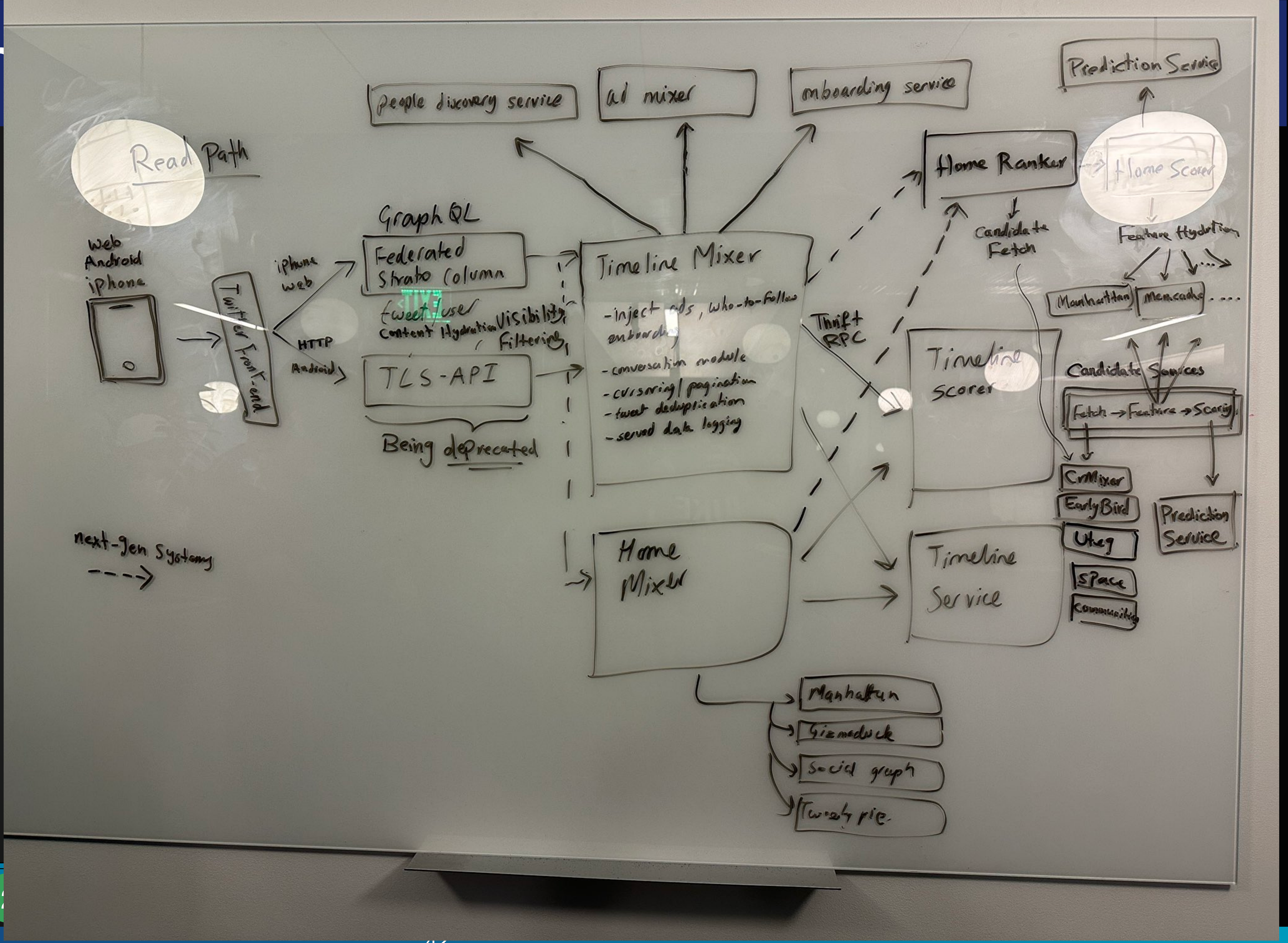
Cor

- Q: What context does this show?



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Context Diagrams

- Context models define the **boundaries** of the system
- They represent key systems that **need to be** developed
- And **the relationship** to the other systems/components

- Also - **what NOT to develop** (to focus investment)

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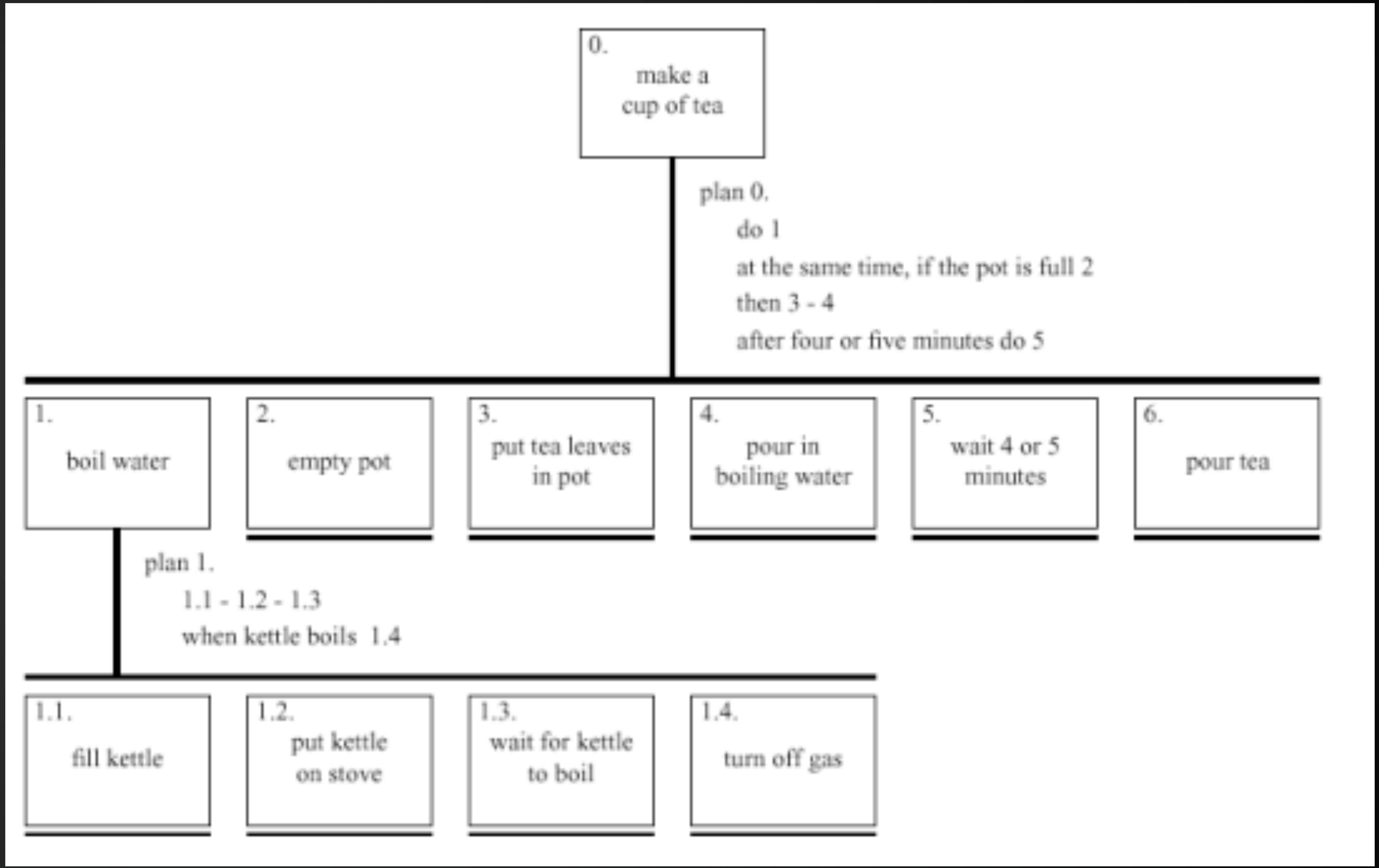
Activity 1 - Context Diagram

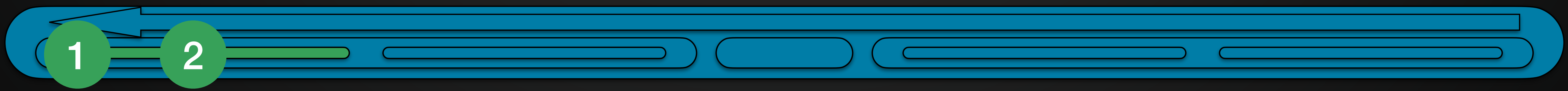
Task Analyses

Hierarchical Task Analysis

- For Identifying subtasks
- Still not good for ‘process’
  - can put notes
  - cant put ‘different actors’ into it
- Nor decision points
  - there is a diagram that can!
- Might make one of these per Use Case?
  - or for when a use case is complex



http://www.comp.lancs.ac.uk/~dixa/papers/HCI-intro-94/



Activity in own time - Task Analysis

Pick a complex use case, and break it down into subtasks

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UML Diagrams

(in Visual Paradigm)

Unified Modeling Language (UML)

- There used to be three popular approaches to producing models
  - ones specifically for modelling object oriented software in software engineering
- In 1990, they got ‘unified’
- UML 2 includes 13 different diagrams
  - each for different purposes
- Its possible to automatically produce outline code from UML
  - called Model Driven Development
  - but this isnt entirely popular - the Somerville book outlines 4 reasons why in Section 5.5

  - https://www.uml-diagrams.org/uml-25-diagrams.html

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UML - 5 Key Diagram Types

- Use Case Diagrams
- Activity Diagrams
- Sequence Diagrams

Behaviour Diagrams
Could be Req, Spec, or Doc

- Class Diagrams
- State Diagrams

Structure Diagrams
of Code - thus more for Spec, and Doc

1 2

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Activity Diagrams

- Used to elaborate workflows for key activities - especially if they involve decisions
- Explains the process, decision points, wait points & parallel work
- Usually to define one Use Case in more detail
- May tie together several Use Cases for one bigger process.

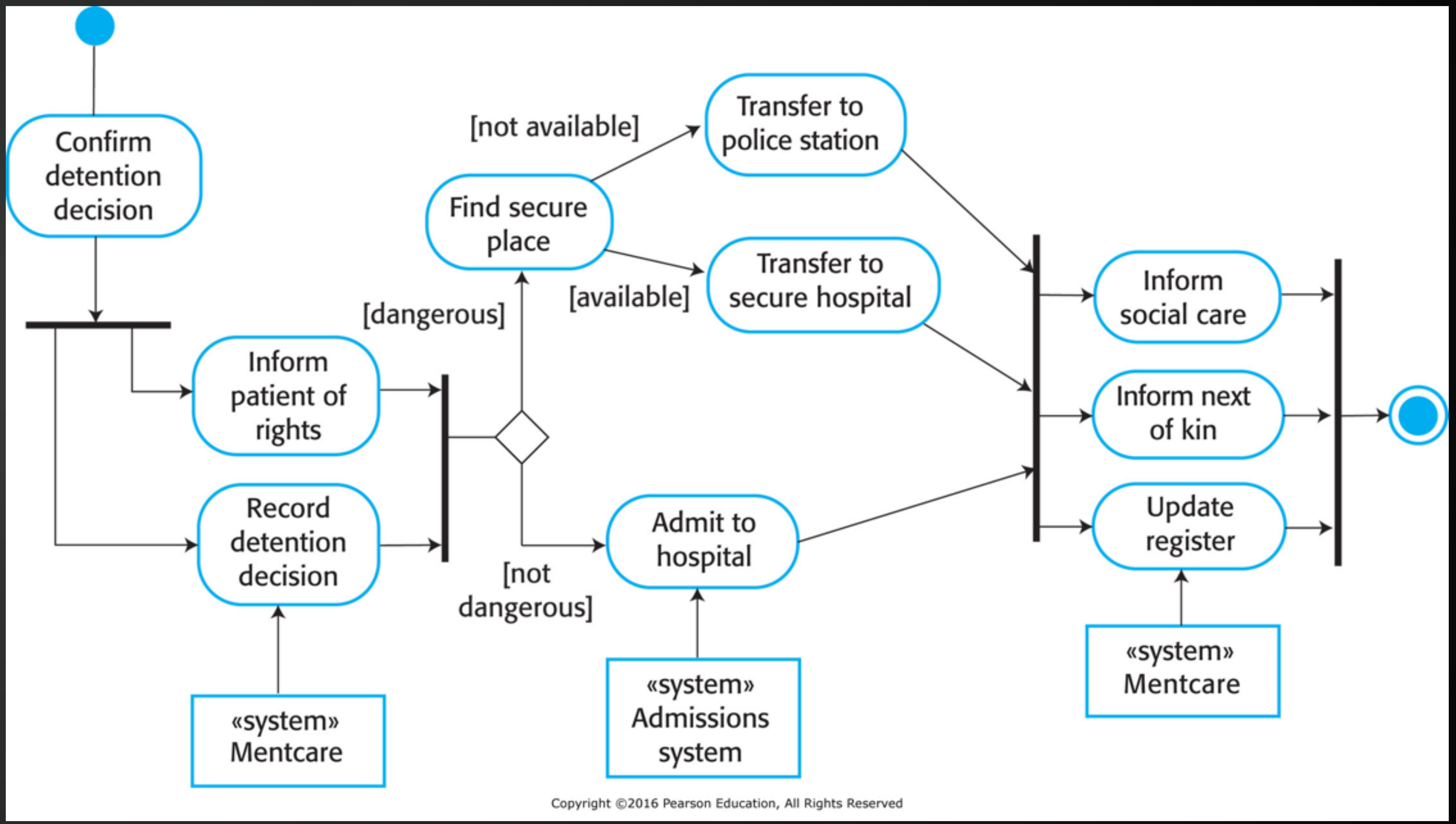
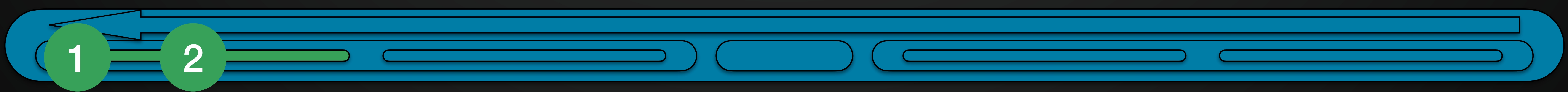
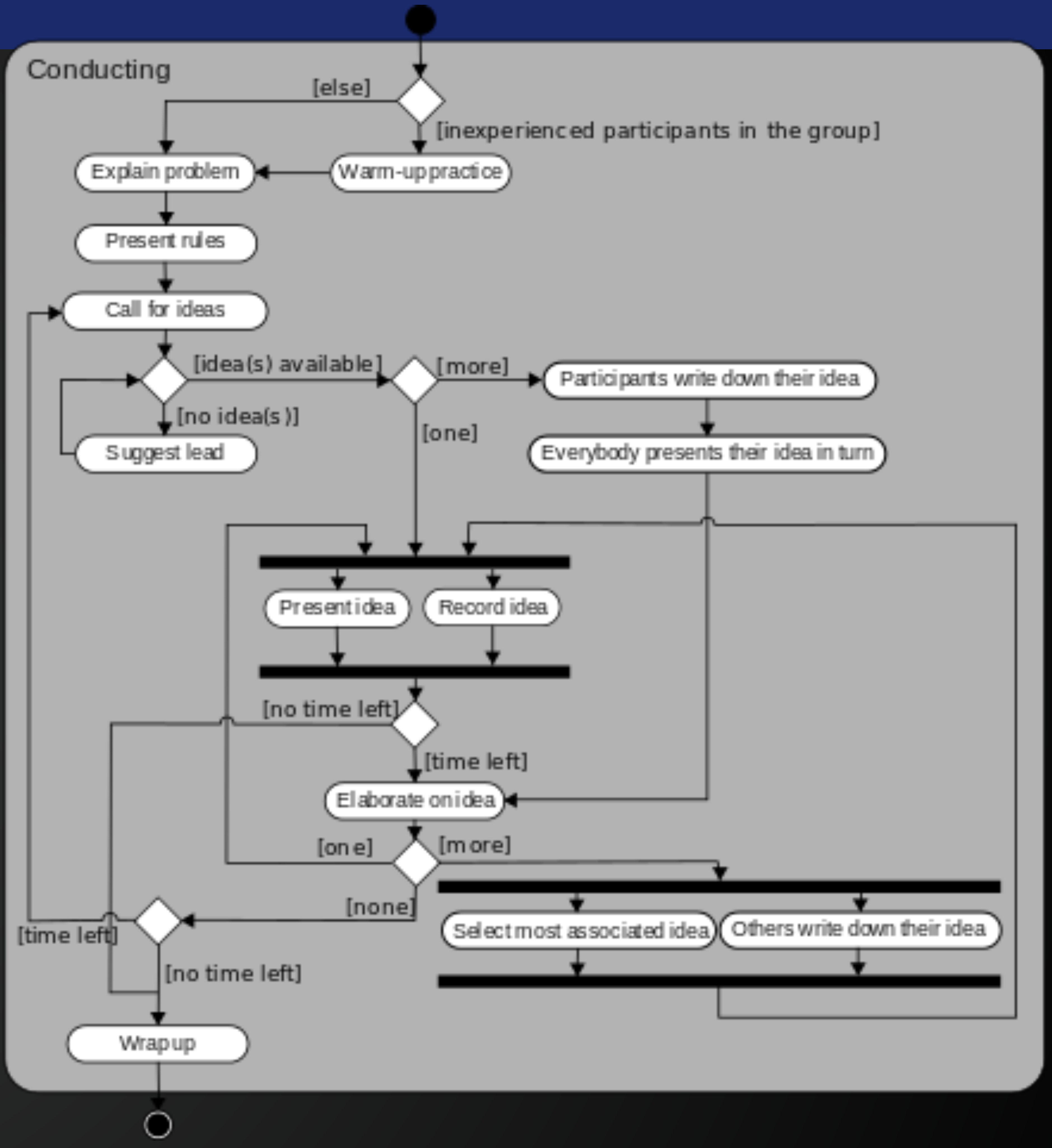


Figure 5.2 *A process model of involuntary detention*



Activity Diagrams

- Rounded rectangles represent actions
- Diamonds represent decisions
- Bars represent the start (split) or end (join) of concurrent activities
- The filled circle represents the start
- The encircled filled circle represents the end



https://en.wikipedia.org/wiki/Activity_diagram

Activity in own time - Activity Diagram

Identify a complex process, involving decisions, from one or more use cases.
Create an Activity Diagram for it.

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UML - 5 Key Diagram Types

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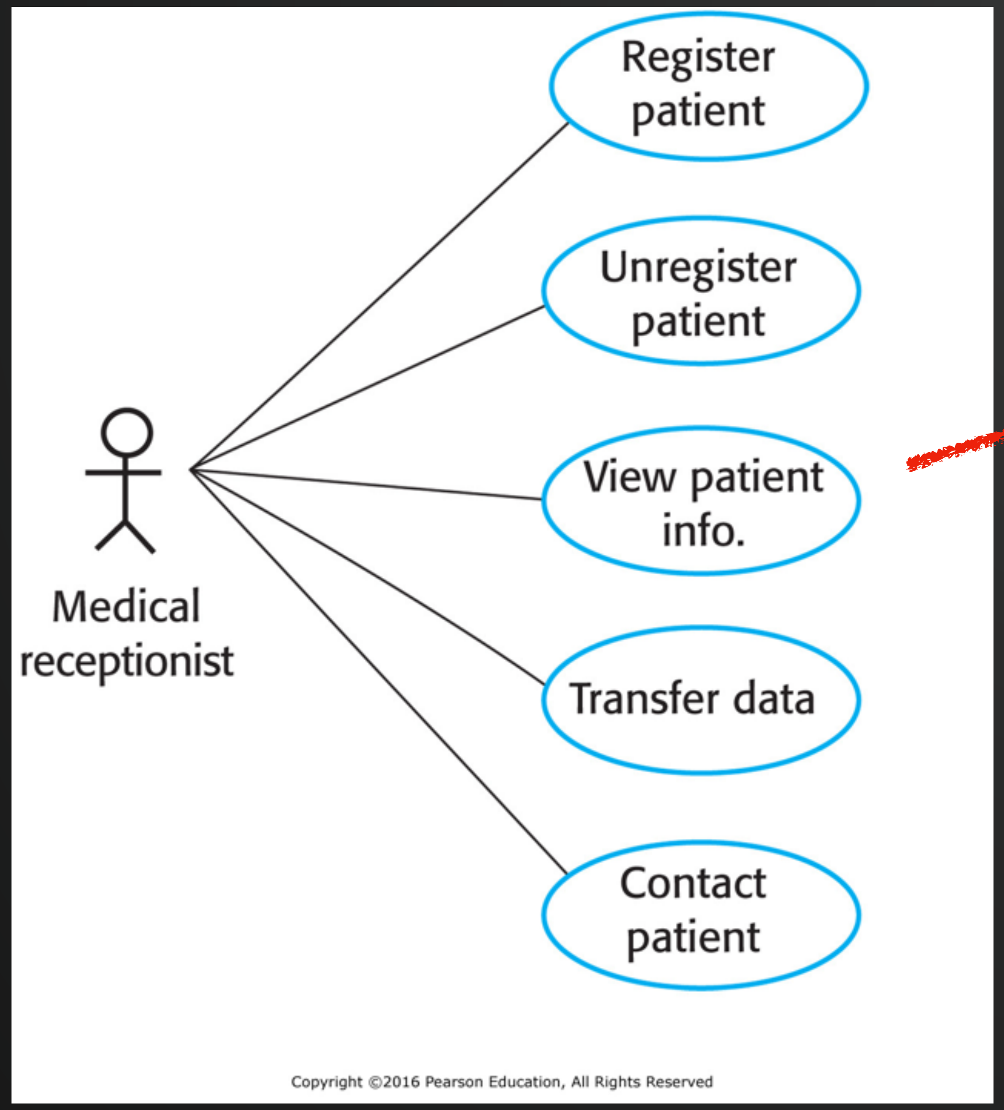
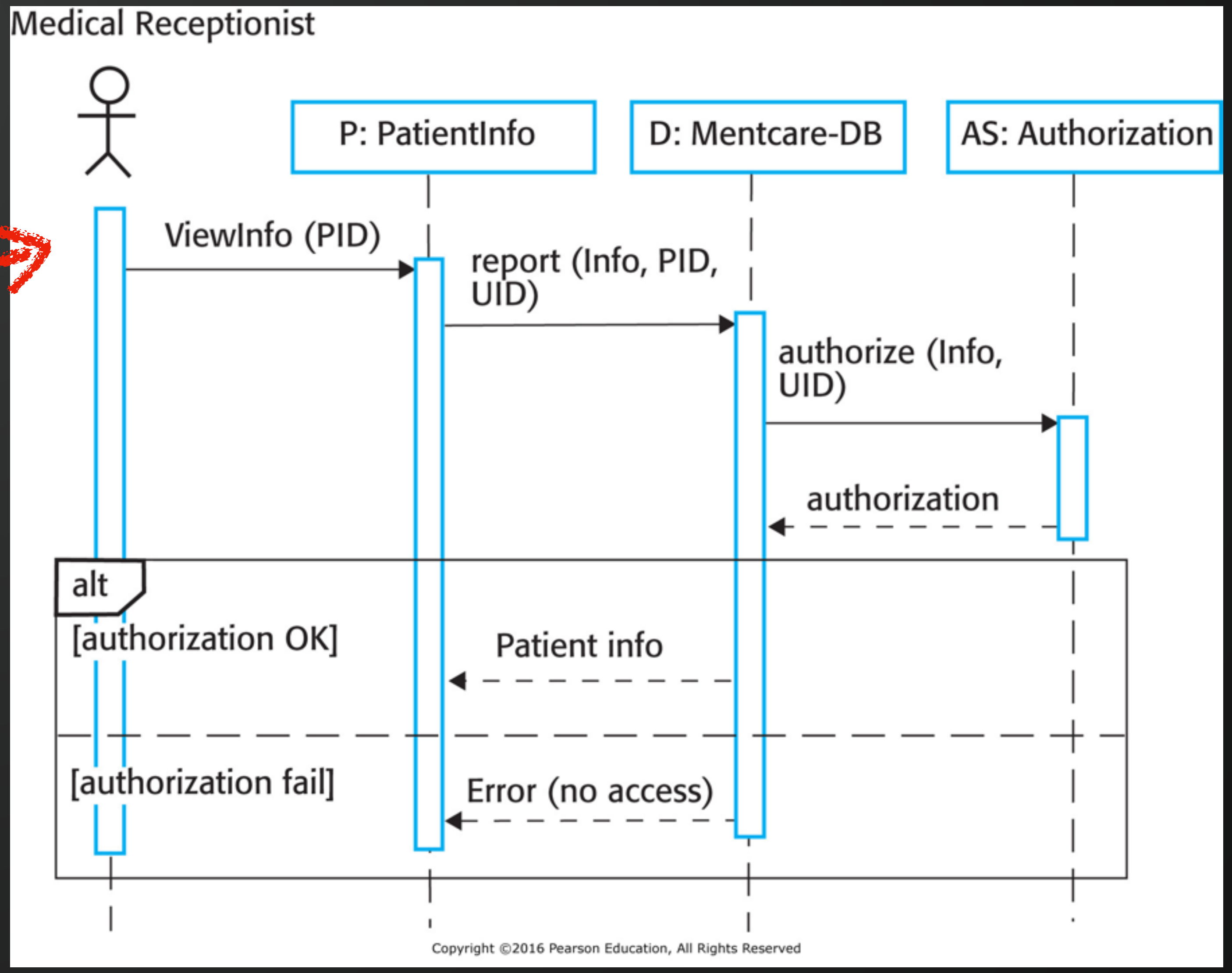
Sequence Diagrams

- Good for complex sharing of information between people and systems
- Could be seen as a series of messages between key components
- Can be (later) used to specify function calls in java objects





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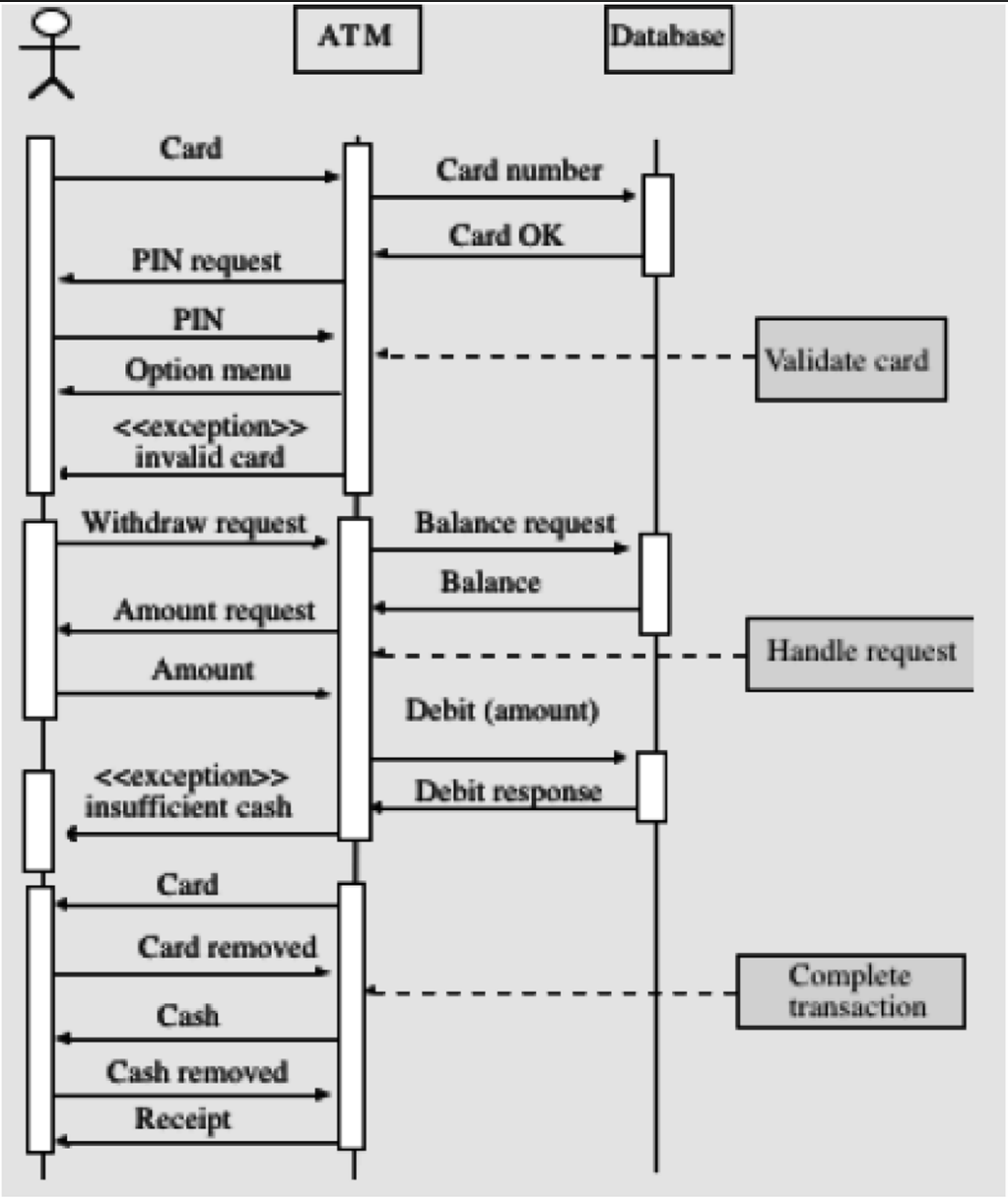
Sequence Diagrams



Source: Sommerville (2004)

Using an ATM

Kevin Curran, David King, (2008) "Investigating the human computer interaction problems with automated teller machine navigation menus", Interactive Technology and Smart Education, Vol. 5 Iss: 1, pp.59 - 79

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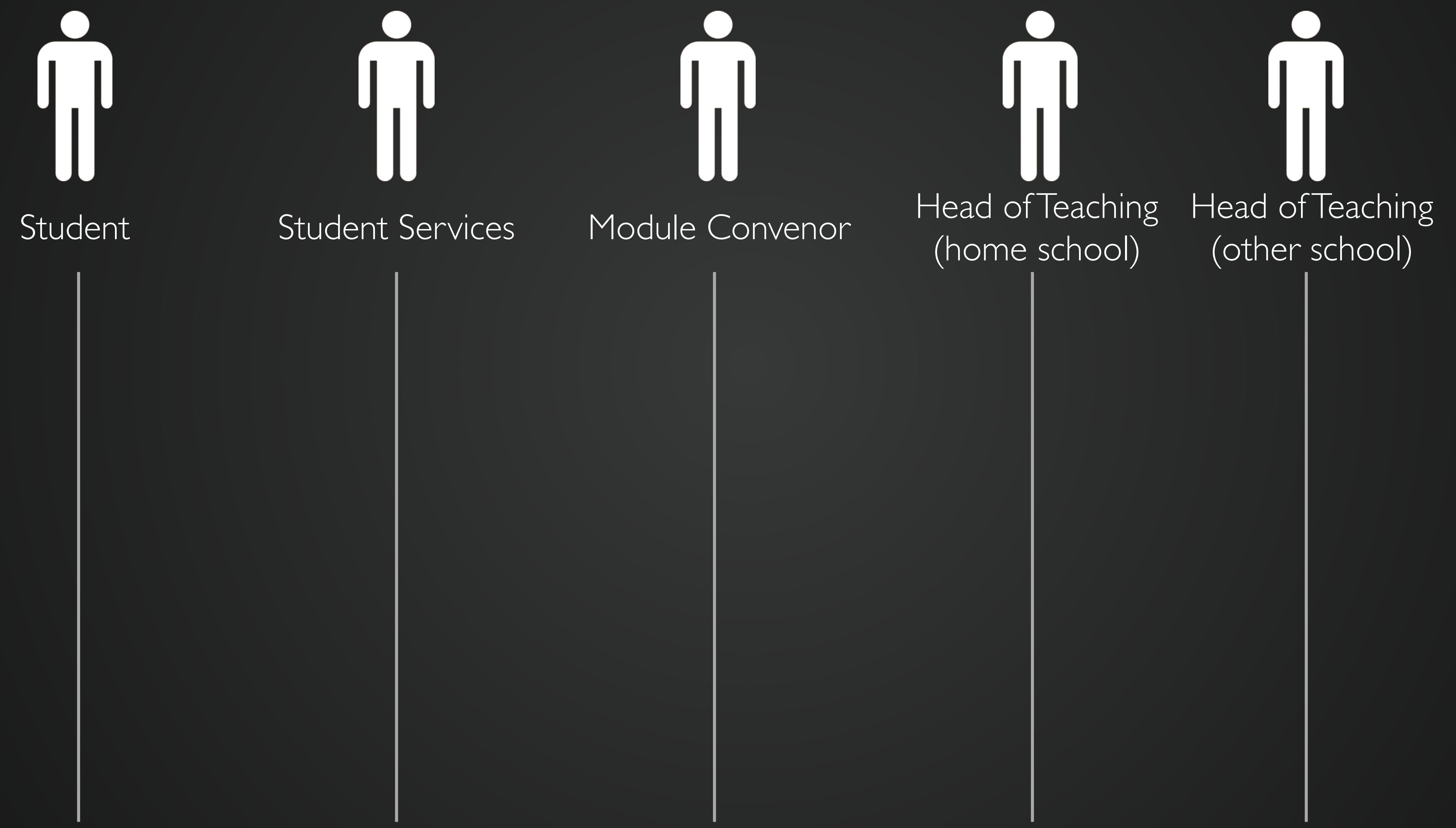
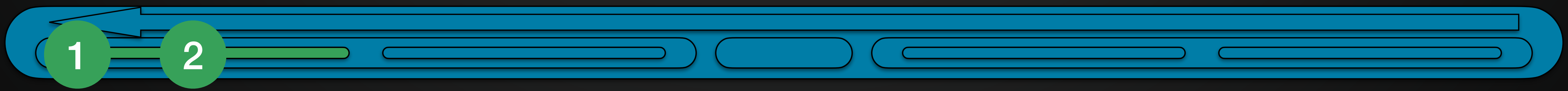
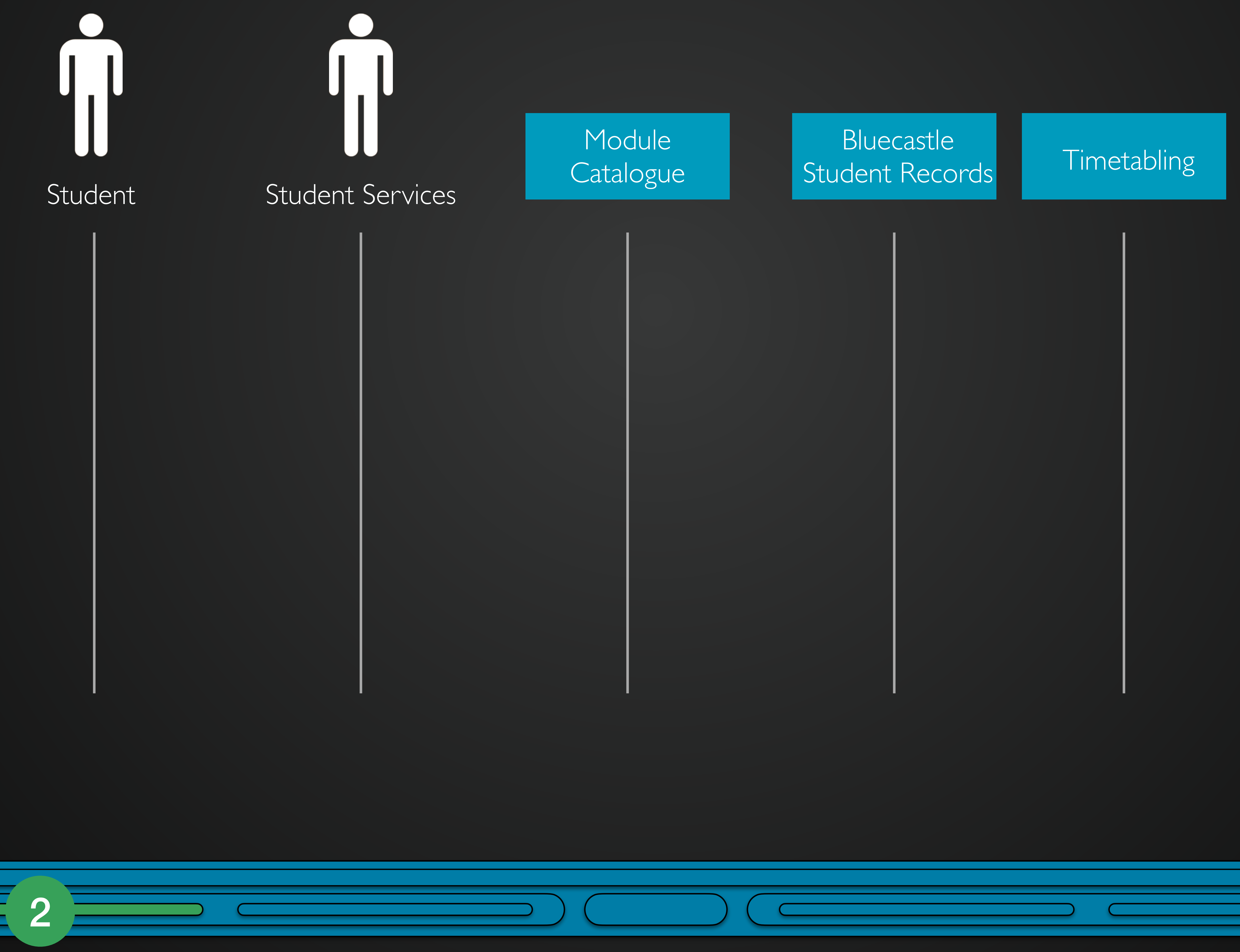
Sequence Diagrams - Activity





Sequence Diagrams - Activity



Other UML
(mostly* for later stages, like specifications)

*but can help in Requirements

UML - 5 Key Diagram Types

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Structure Diagrams
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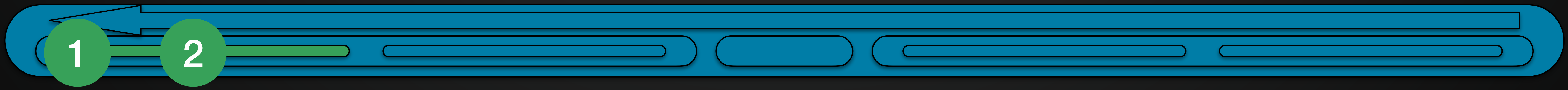

Class Diagrams

- Specify the classes in Object Oriented Code
- A lot like a Entity Relationship Diagram (COMP1004)
- You'll do more of these in OO part of COMP1009
- Done well - this is obviously more like specification
- Could be done at Requirements time
  - a kind of prototype of objects that Req Engineers have found

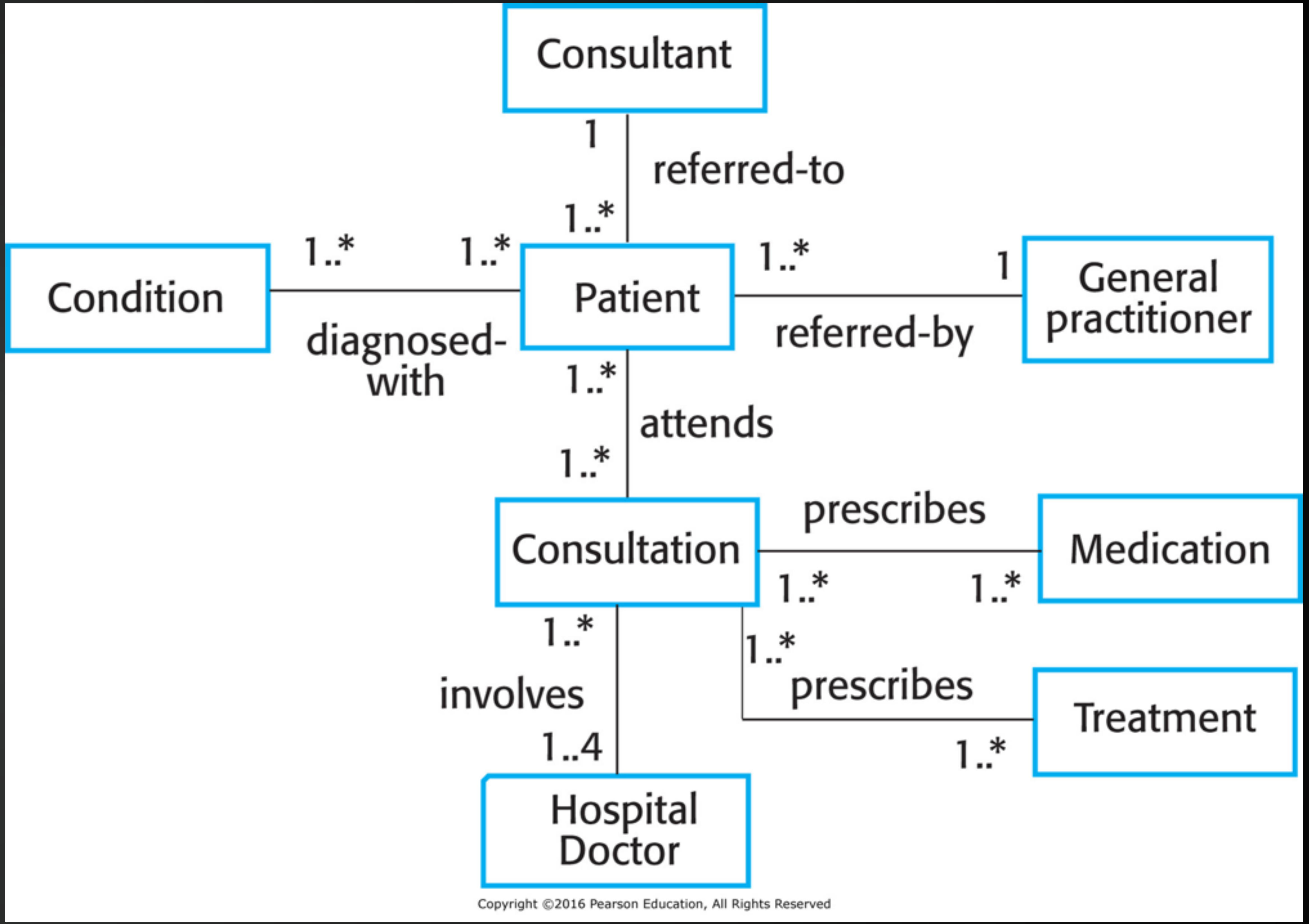

Figure 5.9 *Classes and associations in the Mentcare system*

Class Hierarchy Diagram

- Shows levels of inheritance for classes in Object Oriented code
- Again - likely to revisit in COMP1009

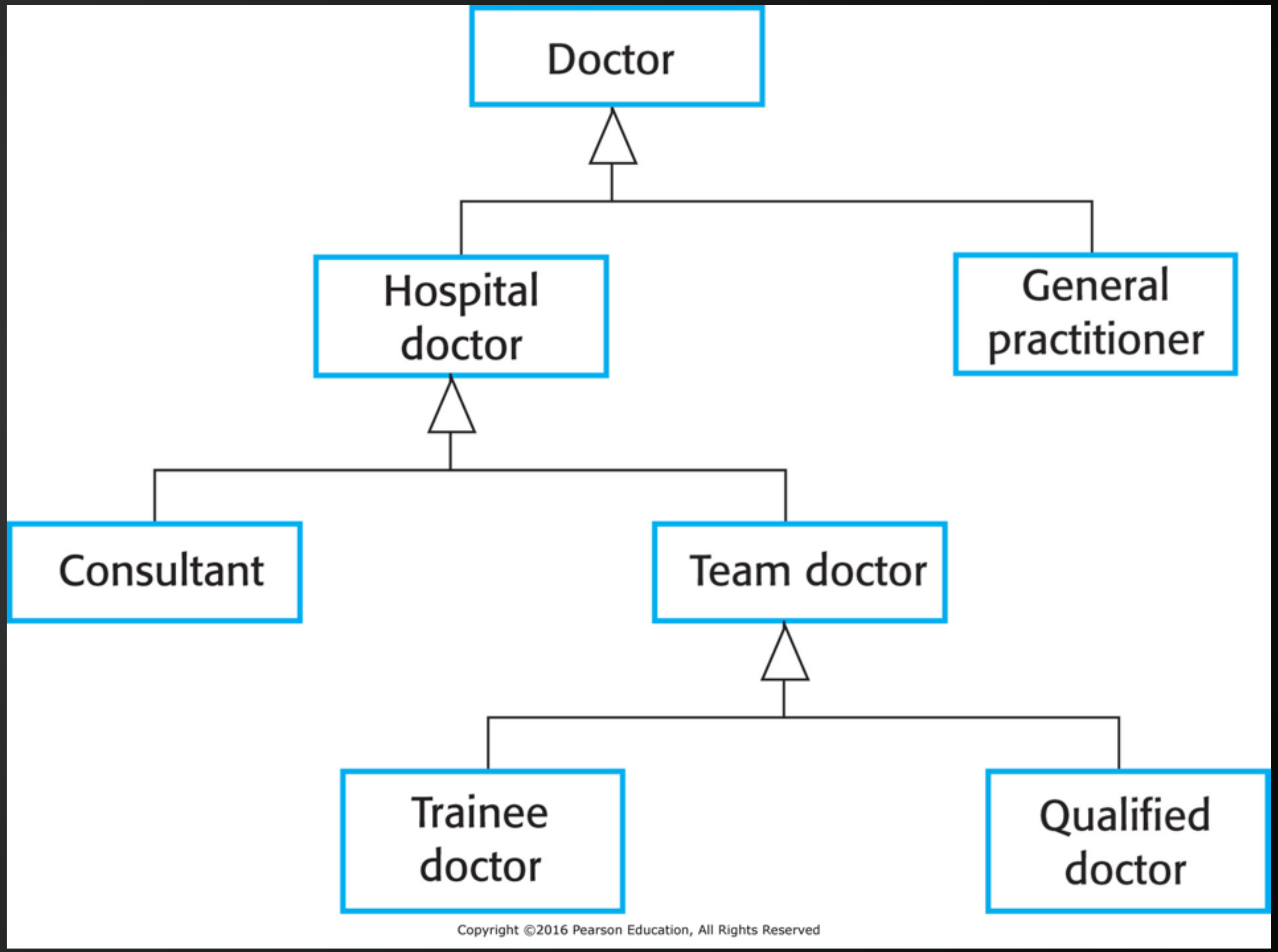

Figure 5.11 *A generalization hierarchy*

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UML - 5 Key Diagram Types

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Structure Diagrams
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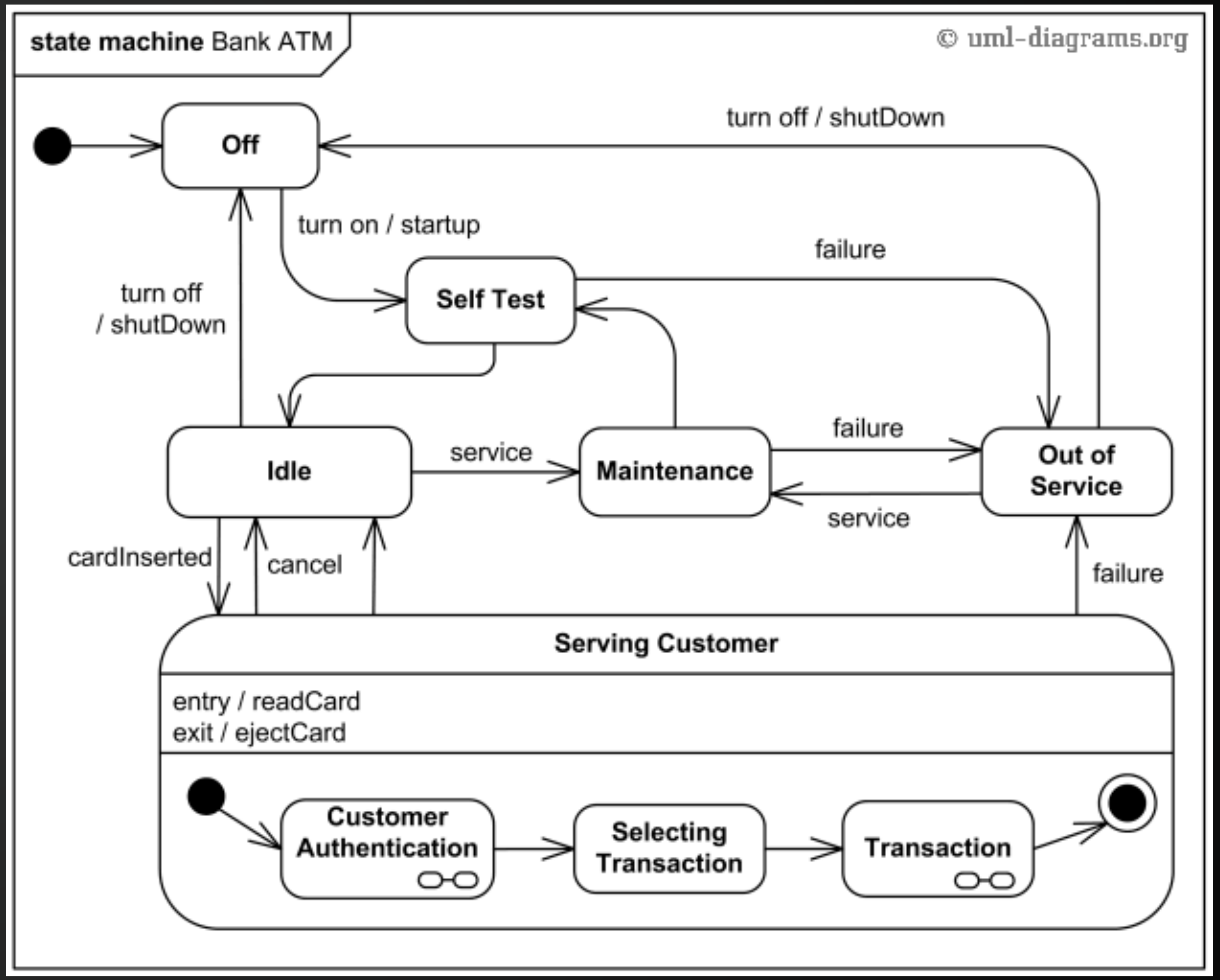
State Diagrams

- Bank Example again
- All the states that ‘**something**’ can be in
- And what actions change it
- Could do this with a ‘module choice form’
- Or ‘student’ going through university
  - applicant
  - successful applicant
  - studying
  - suspended
  - completed



State Diagrams

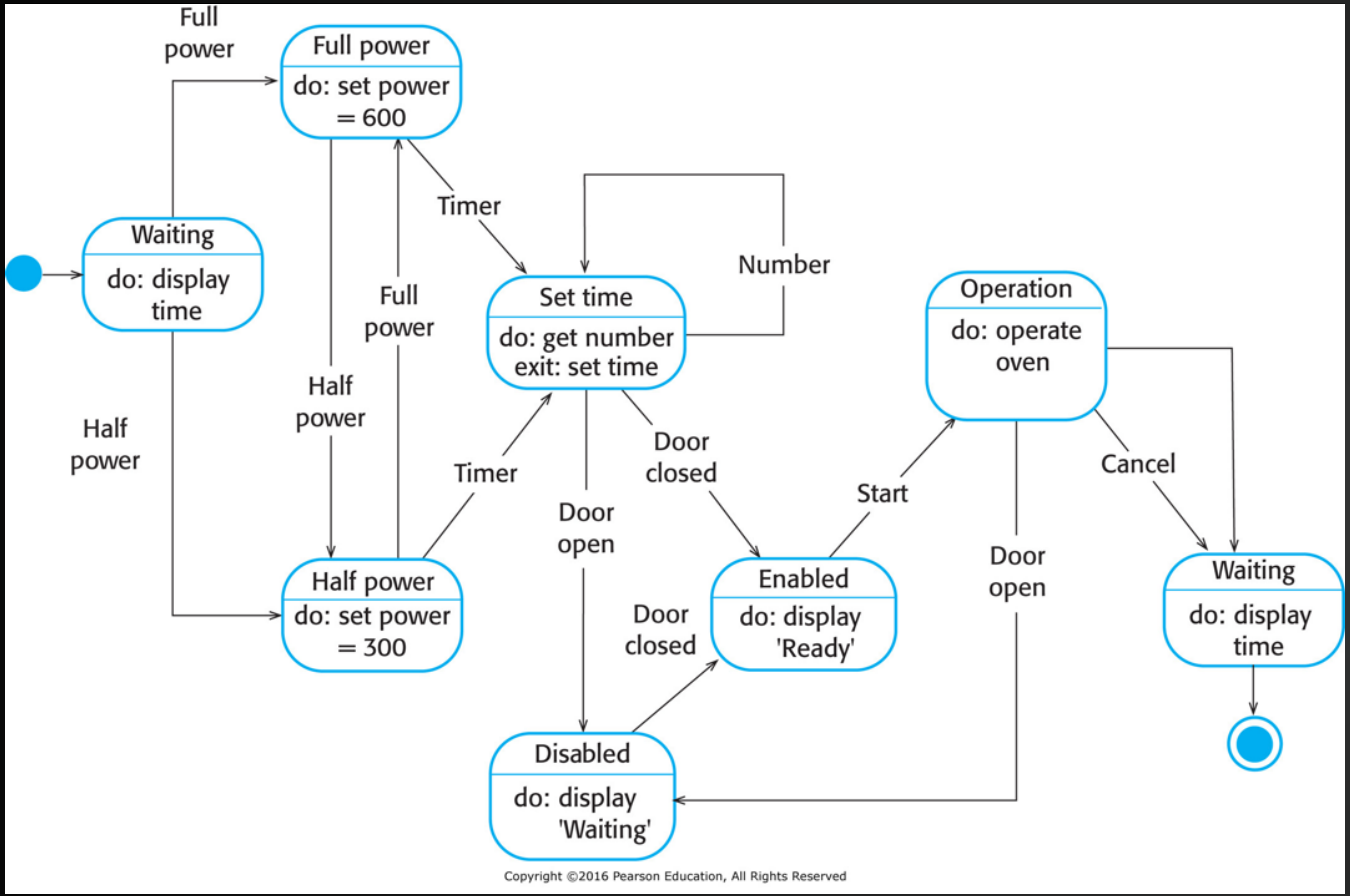


Figure 5.16 *A state diagram of a microwave oven*

State	Description
Waiting	The oven is waiting for input. The display shows the current time.
Half power	The oven power is set to 300 watts. The display shows "Half power."
Full power	The oven power is set to 600 watts. The display shows "Full power."
Set time	The cooking time is set to the user's input value. The display shows the cooking time selected and is updated as the time is set.
Disabled	Oven operation is disabled for safety. Interior oven light is on. Display shows "Not ready."
Enabled	Oven operation is enabled. Interior oven light is off. Display shows "Ready to cook."
Operation	Oven in operation. Interior oven light is on. Display shows the timer countdown. On completion of cooking, the buzzer is sounded for 5 seconds. Oven light is on. Display shows "Cooking complete" while buzzer is sounding.
Stimulus	**Description**
Half power	The user has pressed the half-power button.
Full power	The user has pressed the full-power button.
Timer	The user has pressed one of the timer buttons.
Number	The user has pressed a numeric key.
Door open	The oven door switch is not closed.
Door closed	The oven door switch is closed.
Start	The user has pressed the Start button.
Cancel	The user has pressed the Cancel button.

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Figure 5.18 *States and stimuli for the microwave oven*

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Scenarios

(Not UML)
Bringing it together

Scenarios

"envision and document typical and significant user activities"

– John Carroll, Five Reasons for Scenario-based Design

- Adding Context/Detail to Models/Diagrams
- Not just a story, though
- But a structured description of process

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Scenarios

- Must **define a setting** or context
  - might come from a technology tour (for example)
  - the environment, the things in it, etc
- Must define one or more **actors or users**
  - perhaps a persona
- Must define **goals or objectives**
  - perhaps from a task analysis or user story

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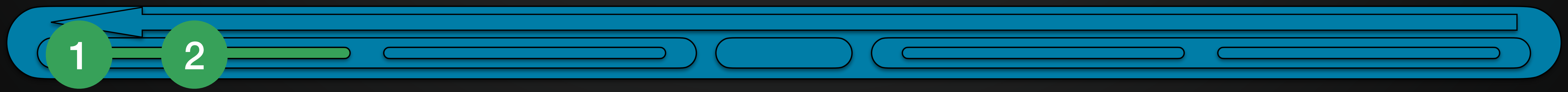
Scenarios

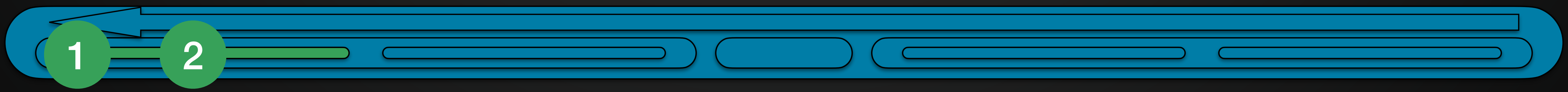
- Must **describe a plot**
  - as a sequence of events
  - perhaps from a task analysis or activity diagram
- **The plot describes how a user, in a context, achieves a goal**
- Scenarios are written text descriptions
  - describing current events
  - OR describing ideal or possible events in design
- Pick ones that *would benefit* from being ‘illustrated’ with detail
  - Scenario for student picking the simple modules?
  - Scenario for student trying to do a 70/50, including a language from another department?
  - Scenario for a module convenor, considering unusual enrolment requests?
  - Scenario for a student services rep processing a series of standard requests?



Scenario Example

- Book recommendations for detail section:
  - the 'normal' flow of events for the scenario
- Book recommendations for notes section:
  - What can go wrong and how to fix
  - Other activities happening in parallel
  - What happens after

Scenario MP3/01	
Title	How does that song go again?
Overview	People: single female, computer literate, works at home Activities: Searching for mp3 tracks Context: Apartment with office/study Technology: Pc
Rationale	An introduction/reasoning for its existence
Detail	Numbered paragraphs in detail describing stage by stage e.g. - P5: She touches the play button and listens. She increases the volume, and the lyrics appear on screen.
Notes	Numbered design issues/questions to be discussed/ considered



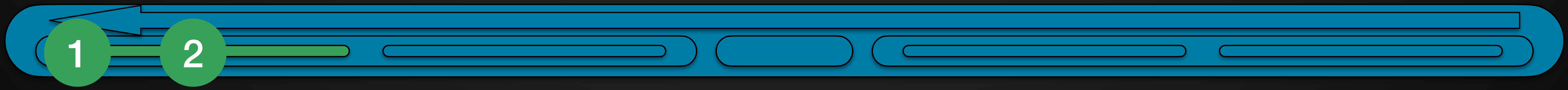
Activity in own time - Scenario

Take an Activity Diagram.
Make a scenario for it in your own time.

Summary

Most Important Thing:

- 1) What do you want to show?
  - a complex decision process
  - a strict sequence of actions
  - subtasks in a complex use case
  - other software to export to, import from, use
  - the structure of a system, etc
- 2) Which diagram or model is best for this?
- Avoid ‘trying to show everything in one diagram’



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- 1) What do you want to show?
  - a complex decision process
  - a strict sequence of actions
  - subtasks in a complex use case
  - other software to export to, import from, use
  - the structure of a system, etc
- 2) Which diagram or model is best for this?
- 3) Write something for each diagram that connects it to other diagrams
  - This is a sequence diagram for User Case X
  - This is an activity diagram that represents the most complex module choice situation, which requires approval at all stages.
- Avoid ‘trying to show everything in one diagram’

Today's Learning Outcomes

- 1) What requirements models are, and why they are useful
- 2) Different types of UML diagrams
- 3) Actually, models aren't just for requirements
  - this is a useful transition point to our next stage

To think about for next lab

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What makes good content?

- What have I told you about doing a good stakeholder analysis?
  - (The answer to this could go in the markdown)

- What have I said makes a good persona?
- What have I told you makes a good use case diagram?
  - What ‘more advanced features’ have I mentioned, and can you use them?

- Producing basic diagrams is like 40% of the way there
- Producing good diagrams that are useful is 70% the way there
- Explaining reasoning for choices and useful insights is the rest